

COMPLETION REPORT



INSTALLATION OF COMPRESSED AIR SYSTEM IN ROOM 601 AND 603

PROJECT STARTED:

FEBRUARY 19, 2024

PROJECT ENDED:

MAY 10, 2024

PREPARED BY:

FOUR BEES ENGINEERING SERVICES
BSME 4-B CTU-Main Campus Batch 2024

SUBMITTED TO:

CEBU TECHNOLOGICAL UNIVERSITY - MAIN CAMPUS
COLLEGE OF ENGINEERING
MECHANICAL AND AEROSPACE ENGINEERING DEPARTMENT



Completion Report

Compressed Air System

I. Project Details

Project Title: Installation of Compressed Air System in Room 601 and 603

Project Start Date: February 19, 2024

Project End Date: May 10, 2024

Total Cost: ₱14,250.00

II. Executive Summary

Installing a compressed air system in a laboratory is essential. It ensures a steady supply of clean air for various tasks, such as powering pneumatic equipment and operating instruments, leading to accurate results. It also enhances safety, saves time, and helps maintain equipment, making it essential for efficient and safe lab operations.

2.1 Project Outcome

The compressed air system greatly benefits students, schools, and laboratory experiments. Students have access to clean air for experiments, ensuring accurate results. Schools benefit from increased safety and efficiency, while experiments are enhanced with consistent air pressure, leading to better outcomes over all.

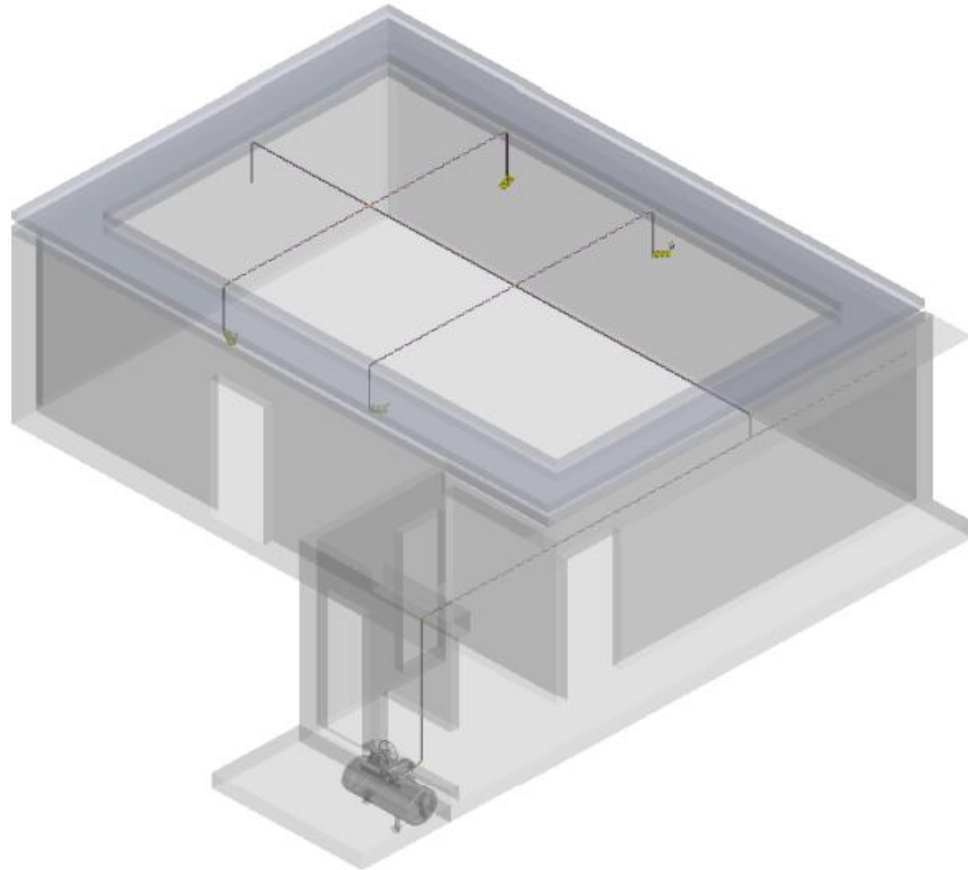
2.2 Project Sustainability

Having a compressed air system in the laboratory ensure reliable results. It provides clean air for accurate experiments and saves time and energy by automating tasks. The system also improves safety, reducing accidents and downtime, making lab operations more sustainable.



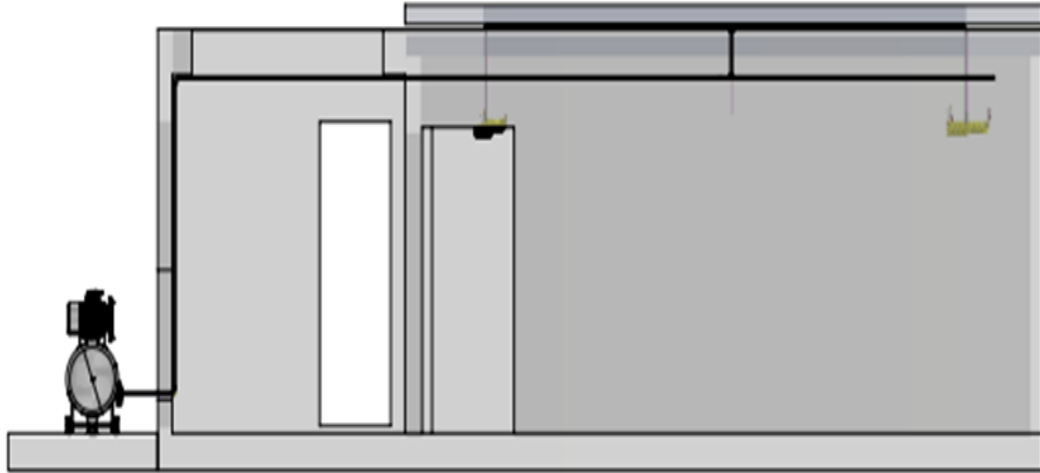
III. Air Compressor Design

Isometric View

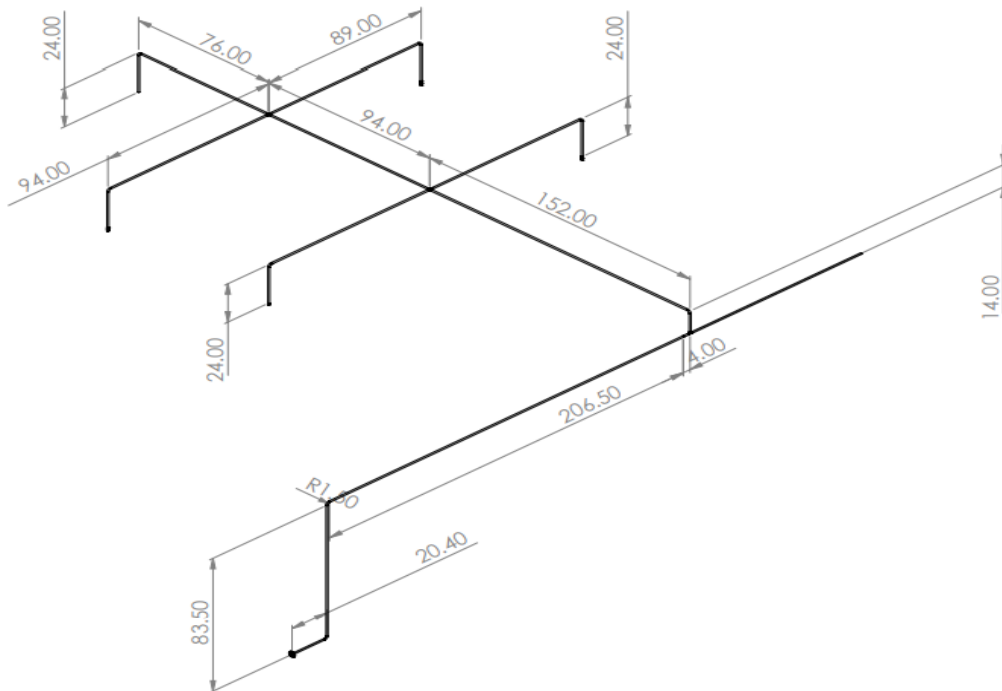




Front View



Drawing Dimension





Given Room Parameters:

Description	Dimension
Room Size	232 x 340 in.
Total Pipe Length (excluding fittings)	1105 in. (28 m) (EN 601)
Total Pipe Length (excluding fittings)	1767 in. (45 m) (Run to EN 603)

Tool Working Pressure:

Demand	Demand Specification	Working Pressure
Air Dust Blowgun	4" Disc	20-50 PSI
Air Angle Grinder	4" Disc, Horizontal	90 – 100 PSI
Air Pencil Grinder	¼" Inlet, General, (DOTCO)	90 PSI
Pneumatic PLC Test Rig	40x100 mm Double Acting Cylinder & Single acting cylinder	7.25 – 14.5 PSI
Working Pressure Range		7.25 – 100 PSI

Calculating Average System Demand:

Type of Tool	Location	Number of Tools (A)	Load Factor (% average load factor) (B)	Tool CFM Per tool when operating (C)	CFM Required if all tools operated simultaneously/ full rated Operation (D)	CFM Required if all tools operated at average load (D)	Total CFM Actually used at average load (AxBxC÷100) (E)	Total CFM Actually used at full rated Operated (F)
[Tool]	[Room No.]	[Quantity]	[CFM]	[CFM]	[CFM]	[CFM]	[CFM]	[CFM]
Air Dust Blowgun	EN601	1	3.0	8.8	8.8	3.0	0.264	0.77
Air Angle Grinder	EN601	2	6.0	23.3	46.6	12.0	2.796	10.86
Air Pencil Grinder	EN601	1	2.30	9.0	9.0	9.0	0.207	0.81
Pneumatic PLC Test Rig	EN603	1	0.53	2.13	2.13	1.13	0.0128	0.024
		1	0.53	2.13	2.13	1.13	0.0128	0.024
Total Quantity		6	Total CFM			26.3 CFM	3.33 CFM	12.5 CFM



Compressor Selector Chart:

Compressor Pressures per square inch		Air Consumption in Cubic Feet Per Minute of Total Equipment		
Cut In	Cut Out	Average Use*	Continuous Operation**	Horsepower of Compressor Required
80	100	Up to – 6.6	Up to – 1.9	1/2
80	100	6.7 – 10.5	2.0 – 3.0	3/4
80	100	10.6 – 13.6	3.1 – 3.9	1
80	100	Up to – 14.7	Up to – 4.2	1
80	100	13.7 – 20.3	4.0 – 5.8	1 1/2
80	100	14.8 – 22.4	4.3 – 6.4	1 1/2
80	100	20.4 – 26.6	5.9 – 7.6	2
80	100	22.5 – 30.4	6.5 – 8.7	2

Air Compressor Specification:

Description	Parameters
Power:	1.5 kW (2 HP)
Voltage:	220-240V, 60Hz
Speed:	1700 RPM
Displacement:	93 L/min (3.28 CFM)
Max. Pressure:	8bar (116 Psi)
Tank Capacity:	24L(6.3Gal)
Discharge:	¼ in.



Total TC120246P 2HP Direct Couple Air Compressor
Pump lubricated with oil
Packed by color box

Link: <https://shop.goldpeaktools.com.ph/products/total-tc120246p-2hp-direct-couple-air-compressor>



Pipe Sizing

SCFM Air Flow	25	50	75	100	150	200	300	500	1000	Compressor HP
4	1/2	1/2	1/2	1/2	1/2	1/2	1/2	3/4	3/4	1
12	1/2	1/2	1/2	3/4	3/4	1/2	3/4	1	1	3
20	3/4	3/4	3/4	3/4	1	3/4	1	1 1/4	1 1/4	5
30	3/4	3/4	1	1	1	1	1 1/4	1 1/4	1 1/4	7 1/2
40	3/4	1	1	1	1 1/4	1	1 1/4	1 1/2	1 1/2	10
60	1	1	1 1/4	1 1/4	1 1/4	1 1/4	1 1/2	1 1/2	2	15
80	1	1 1/4	1 1/4	1 1/4	1 1/2	1 1/4	1 1/2	2	2	20
100	1 1/4	1 1/4	1 1/2	1 1/2	1 1/2	1 1/2	2	2	2 1/2	25
120	1 1/4	1 1/2	1 1/2	1 1/2	2	1 1/2	2	2 1/2	2 1/2	30
160	1 1/4	1 1/2	1 1/2	2	2	1 1/2	2 1/2	2 1/2	3	40
200	1 1/2	2	2	2	2	2	2 1/2	3	3	50
240	1 1/2	2	2	2	2 1/2	2	2 1/2	3	3	60
300	2	2	2	2 1/2	2 1/2	2	3	3	3 1/2	75
400	2	2 1/2	2 1/2	2 1/2	3	2 1/2	3	3 1/2	4	100
500	2	2 1/2	2 1/2	3	3	2 1/2	3 1/2	3 1/2	4	125

Pipe size: 1/2 in."

WL 102. This size chart lists the IPS (Iron Pipe Size) pipe sizes and pressure class ratings for our HDPE 4710 pipe with IPS sizes ranging from 3/4" to 54".

Link: <https://wlplastics.com/hdpe-pipe-sizes/>



Valve & Fittings

Valve & fittings	Quantity	Pipe Size	Le	L_{TE}
Ball Valve	1	½"	6"	6
90 deg. Elbow	8		5"	40"
Three-way/ Tee	1		7.5"	7.5"
Four-way/ Cross	2		10"	20"
Pneumatic Air Nipple	5		3"	15"
End Cap	1		2"	2"
Total Estimated Length of Valve and Fittings				90.5"

IV. Bill of Materials

ITEM NUMBER	ITEM NAME	DESCRIPTION	QUANTITY	UNITS	UNIT COST	TOTAL PART COST
1	AIR COMPRESSOR	2 HP, 24L	1	1	P7,000.00	P7,000.00
2	PIPE	- HDPE, 1/2", SDR 9	45 m	1 m	P35.00	P1,238.00
3	ELBOW	PE COMPRESSION FITTING, 1/2"	7	1	P56.00	P392.00
4	TEE	- COMPRESSION FITTING, 1/2"	1	1	P63.00	P63.00
5	CROSS	- COMPRESSION FITTING, 1/2"	3	1	P91.00	P273.00
6	END CAP	COMPRESSION FITTING, 1/2"	1	1	P27.00	P27.00
7	BALL VALVE	- COMPRESSION FITTING, 1/2"	1	1	P125.00	P125.00
8	PNEUMATIC AIR NIPPLE	1/2"	4	1	P99.00	P396.00
9	AIR DUST BLOW GUN	4" DISC, 20-50 PSI	1	1	P150.00	P150.00
10	AIR ANGLE GRINDER	4" DISC, HORIZONTAL, 90-100 PSI	1	1	P1,000.00	P1,000.00
11	AIR PENCIL GRINDER	1/4" INLET, GENERAL, (DOTCO), 90 PSI	1	1	P2,500.00	P2,500.00
12	PRESSURE REGULATOR		1	1	P1,086.00	P1,086.00
						P14,250.00



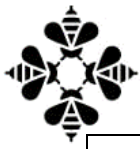
JIT CHART	42		Documentation	61 days	Mon 19/02/24	Mon 13/05/24	1	
	43		Business Plan					
	44		Procedures					
	45		Completion Report					



Compressed Air System Test Form

Project Information
Project Name:
Location:
Date:

Sr.	Description	Yes	No	N/A	Comments
1.	Is there a leakage in the HDPE Pipe?				
2.	Is the compressor generating too much noise and vibration?				
3.	Do fittings bridge the gaps and provide secure connections, ensuring a continuous system free of leaks?				
4.	Meter gauge working properly.				
5.	Exhaust not directed inside any room. (Internal Combustion Air Compressors shall be kept outside in the well-ventilated area).				
6.	Air hoses in good condition without cracks/damages.				
7.	Fire extinguisher provided.				
8.	Can clamps properly secure the pipe by using the self-tapping screw, and can they be tightly installed?				
9.	Are pipe marking color codes applicable for identifying pipes in air systems?				
10.	Can air compressors operate efficiently?				
11.	Any damage/cut pipe/visible damage.				
12.	Have both the intake and discharge valves of the				



	compressor been checked for signs of damage, wear, or blockage?				
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Remarks :

	Date	Signature
Inspected by:		
Project Manager:		
Test Engineer:		

VI. Documentation

During the installation of a compressed air system



Acceptance Form
Compressed Air System

We are formally submitting our completion report as conclusive evidence of the successful completion of our project.

Date: _____

Dr. RONALD M. GALINDO

Dean-College of Engineering

ENGR. MARLITO V. PATUNOB

ME Chairman

